

WHAT CAUSES EMOTIONS?

Life is an emotional roller coaster. In the course of a single day, we can experience moments of happiness, surprise, anger, fear, and sadness. These are instinctive and sometimes fleeting responses to our varying thoughts and experiences. Recalling a special moment from childhood can trigger feelings of joy and nostalgia, while the loss of a loved one can bring overwhelming sadness and despair.

COME ONE! COME ALL!

PRIMARY EMOTIONS

In the middle of the human brain is the limbic system, which produces our emotions. At its center is the amygdala, a small nut-shaped mass believed to be responsible for our main emotions. Psychologists think that people have six primary emotions. These emotions are so strong that they are difficult for people to mask, so facial expressions reveal a lot about our feelings.

FEAR

Look at this scaredy-cat. He's so afraid of the ringmaster that fear is written all over his face. Fear exposes the whites of the eyes, while blood drains from the face, making it very pale. The mouth may open in horror.

TEMPER TANTRUMS

While primary emotions come from the amygdala in the limbic system, other parts of the brain serve to protect us by hiding and controlling our true emotions from other people. These are called the frontal lobes, and they take years to develop, finally reaching full capacity during our 20s. This is why children are more likely to have temper tantrums. Their amygdalas are producing strong primary emotions, but their frontal lobes are not yet ready to prevent them from reacting badly.

"LITTLE EMOTIONS ARE THE GREAT CAPTAINS OF OUR LIVES, AND WE OBEY THEM WITHOUT REALIZING IT."

Vincent van Gogh (1853–1890), Dutch painter

ANGER

Don't get on the wrong side of this ringmaster. Note how red he is—blood is rushing to his head in a typical display of anger. The eyes narrow and muscles pull the eyebrows down and in, causing wrinkles to appear above them. The mouth closes tightly or snarls with fury.

SURPRISE

Random acts or unexpected news can cause surprise. Facial responses may be similar to fear, though there are differences. The mouth drops open and the eyes widen. While fear can be disguised, surprise is difficult to mask, as it is so spontaneous.

JOY

The most obvious expression of joy is a genuine smile. It lifts the cheeks and crinkles the skin around the eyes. Teeth are exposed as the mouth opens. The smile is transmitted to the brain, where joyful feelings become heightened.